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MILITARY SPECIFICATION

DETECTOR-DEWAR DT-591A/UA

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of common module, Detector-Dewar DT-591A/UA.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-726	- Packaging Requirement Codes
MIL-STD-756	- Reliability Modeling and Prediction
MIL-STD-810	- Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-5855

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

SPECIFICATIONS

B2-2303020104A	- Cooler, Cryogenic, Mechanical Stirling, (HD-1132/UA)

DRAWINGS

USA CECOM

54490-5004385	- Detector-Dewar DT-591A/UA
80063-SM-C-969732	- Bellows

2.2 Non-Government documents. The following document, of the exact issue shown, forms a part of this specification to the extent specified herein. In the event of conflict between the document referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

Standard Procedures for Testing Infrared Detectors and Their Performance, R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Detector-Dewar DT-591A/UA, referred to herein as the Detector-Dewar, is an opto-electronic device sensitive to radiation in the 7.5 to 12.0 micrometer spectral region. The Detector-Dewar shall use a 60-element detector array of photoconductive mercury-cadmium-telluride operated from a supply of +3.0, ±0.2 volts direct current (Vdc). The detector bias resistors shall be an integral part of the Detector-Dewar.

3.2 Construction. The Detector-Dewar shall be constructed in accordance with 5004385 and as specified herein.

3.2.1 Weight. Weight of the Detector-Dewar shall not be greater than 0.50 pound including bias resistors.

3.3 First article. When specified in the contract (see 6.2) the contractor
shall furnish first article units for inspection (see 4.3).

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings.

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3.5 Components. The Detector-Dewar shall consist of a 60-element detector array and a resistor bias pack. The Detector array orientation with respect to the Dewar mounting flange shall be variable through 360°. At least two reusable, electrically activated getters shall be an integral part of the Detector-Dewar for periodic maintenance of the vacuum, and each getter shall withstand at least seven firings after shipment. The getter activation current (amperes) and time (minutes) shall be provided on the label specified in 5004385.

3.6 Performance characteristics.

3.6.1 Detector-Dewar defects. Unless otherwise specified herein, the Detector-Dewar shall be operated at 80 (+5, -20) K. The Detector-Dewar shall contain no more than two (2) defective elements, as defined in 6.3. Elements 85 to 96 shall be defect-free. Defects shall be separated from one another by at least four (4) nondefective elements. Elements which are open (dead) or have a D^*BB of less than 5.0×10^9 shall be electrically strapped to an adjacent element with common ground.

3.6.2 Time constant. The time constant of each detector element shall not be greater than 2 microseconds.

3.6.3 Sensitive area. The sensitive area of the detector elements shall fall within the dimensions specified on 5004385.

3.6.4 Crosstalk. The crosstalk signal level between any two elements shall be less than -26 dB.

3.6.5 Blackbody detectivity. The blackbody detectivity of all nondefective elements shall not be less than 7.5×10^9 cm $Hz^{\frac{1}{2}}$ w-1 except for channels 85-96 which shall not be less than 9.0×10^9 cm $Hz^{\frac{1}{2}}$ w-1 when the Dewar is cooled with LN_2 to 77 $\pm$ 3 K.

3.6.5.1 Noise. The detector noise of each element shall be not less than 2.5×10^{-9} volts rms $Hz^{\frac{1}{2}}$.

3.6.6 Detector resistance. The detector resistance of each element shall be not less than 20 ohms and not greater than 150 ohms.

3.6.7 Responsivity. The average 500 K blackbody responsivities of the acceptable elements shall not be less than 8.3×10^3 volts/watt nor greater than 20×10^3 volts/watt when the Dewar is cooled with LN_2 to 77, $\pm$3 K. The blackbody responsivity of each element shall not vary by more than $\pm$20 percent of the mean value of the elements which are not defects.

3.6.8 Temperature sensors. At least two calibrated 2N2222 temperature sensor chips shall be installed at or near the detector array location in the Dewar to provide an electrical signal which is an analog of temperature. The temperature sensor shall have a junction voltage of 1.060, $\pm$ 0.015 volt with a 1.00 ma $\pm$0.01 ma forward current when the Detector-Dewar is operated at 77, $\pm$ 3 K. The voltage shall be recorded on the appropriate label on the Detector.

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3.6.9 Spectral response. Each element of the Detector-Dewar shall have a spectral sensitivity of 7.7, ±0.3 to 11.7, ±0.5 micrometers.

3.6.10 Detector bias. The average bias current of the array shall not exceed 3.0 milliamperes per element, and no one element shall take a bias current of more than 4.0 milliamperes. The total bias power dissipated by the array shall not exceed 30 milliwatts. The detector element shall be biased with current flow in the same direction. The detector bias pack shall be operated at +3.0V ± 0.2Vdc.

3.6.11 Dewar warm and cold shields. The incorporation of an internal cold shield is optional. If incorporated, the Detector-Dewar shall meet the following field of view (F.O.V.) requirements: The minimum half angle F.O.V. from the center of the end elements of the array in the elevation direction shall be 14 degrees. In the direction perpendicular to the array's axis, the cold shield(s) shall allow for each element, no vignetting of any ray to any point on the detector surface within an angle of 17 degrees measured from and on either side of a line perpendicular to the array's axis and parallel to the optical axis.

3.6.12 Heat load. The detector heat load shall not exceed 120 milliwatts when all 60 detector elements are operating at their final bias current and the Dewar is cooled to 80, ±5 K.

3.6.13 Vacuum integrity. The Dewar shall comply with heat load requirements of 3.6.12 for a period of at least one (1) year.

3.6.14 Microphonics - When cooled to a temperature of 60-80 K, biased in accordance with paragraph 3.6.10, and vibrated in accordance with paragraph 4.6.14, all elements of the Detector-Dewar shall have no noise peaks above the white noise reference value greater than 6dB from 300 to 10,000 Hz, and 10dB from 10,001 to 20,000 Hz.

3.6.15 Apparent thermal mass (cooldown) - When mated to a cryogenic cooler HD-1132/UA capable of cooling a test mass of 1.25 grams (250 joules) of aluminum (figure 3) from ambient (296 K) to 80 K in not less than 5 nor greater than 6 minutes, the dewar cooldown time from ambient temperature to 80 K shall not exceed the time achieved with the test mass.

3.7 Environmental conditions.

3.7.1 Temperature shock. The Detector-Dewar shall not be damaged (see 6.3.6) by sudden changes in temperature between -54°C and +71°C.

3.7.2 High temperature. The Detector-Dewar shall not be damaged by storage or operation to +71°C.

3.7.3 Low temperature. The Detector-Dewar shall not be damaged by operation to -54°C or storage to -62°C.

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3.7.4 Shock. The Detector-Dewar shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.7.5 Vibration. The Detector-Dewar shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in figure 1.

3.8 Reliability. The Detector-Dewar shall have a specified mean time between failure (MTBF) of at least 27,600 hours.

3.9 Treatment and painting. Unless otherwise specified, the Detector-Dewar shall be treated and painted in accordance with the applicable drawing.

3.10 Preconditioning. Each Detector-Dewar shall be subjected to random vibration and temperature cycling prior to group A testing as specified in figure 2.

3.11 Nameplates and product marking. Unless otherwise specified, the Detector-Dewar module, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection before packaging (see 4.5).

c. Packaging inspection (see 4.9).

4.3 First article inspection. When specified in the contract, the first article inspection shall be performed by the contractor on a minimum of 5 preconditioned (see 4.10) sample Detector-Dewars.

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4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Weight not as specified	2.5	3.2.1	MIL-STD-252
Marking not as specified	2.5	3.11	
Workmanship not as specified	2.5	3.12	MIL-STD-454
Treatment and painting not as specified	2.5 1/	3.9	5004385
Dimensions not as specified	2.5	3.2	5004385
Getters and label not as specified	1.0	3.5	5004385
Detector orientation range not as specified	1.0	3.5	5004385
Sensitive area not as specified	2/	3.6.3	5004385
Dewar warm and cold shields not as specified	2/	3.6.11	5004385

1/ MIL-STD-105, Inspection Level S3 or objective quality evidence.
2/ Inspect every array 100 percent and inspect at least 2 elements in each array.

4.3.2 Tests. First article inspection shall be conducted upon units which have successfully completed the inspections specified in 4.3.1. Tests may be conducted in any order, except that microphonics shall be performed after shock and vibration (4.7.4 and 4.7.5). Failure of any tests shall constitute first article failure.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Detector-Dewar defects	3.6.1	4.6.1
Time constant	3.6.2	4.6.2
Crosstalk	3.6.4	4.6.4
Detectivity	3.6.5	4.6.5
Detector resistance	3.6.6	4.6.6
Responsivity	3.6.7	4.6.7
Temperature sensors	3.6.8	4.6.8
Spectral response	3.6.9	4.6.9
Detector bias	3.6.10	4.6.10
Heat load	3.6.12	4.6.12
Vacuum integrity	3.6.13	4.6.13
Microphonics	3.6.14	4.6.14
Apparent thermal mass (Cooldown)	3.6.15	4.6.15
Temperature shock	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.2
Shock	3.7.4	4.7.4
Vibration	3.7.5	4.7.5
Reliability	3.8	4.8

4.3.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.h).

4.4 Inspection procedures. Unless otherwise specified herein, the Detector-Dewar shall meet the specified performance requirements at an ambient temperature of +23°C, ±5°C, when operated at a temperature of 80, ±5 K, and biased to +3.0, ±0.2 Vdc.

4.5 Quality conformance inspection before packaging.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on all (100%) of the Detector-Dewars following successful inspection of 4.5.1 and preconditioning of 4.10. Group A tests listed in table III may be performed in any order except that Detector-Dewar defects (4.6.1) shall be run last. Failure of any test shall be cause for rejection of that unit.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Microphonics	3.6.14	4.6.14
Detectivity	3.6.5	4.6.5
Detector resistance	3.6.6	4.6.6
Responsivity	3.6.7	4.6.7
Temperature sensors	3.6.8	4.6.8
Detector bias	3.6.10	4.6.10
Heat load	3.6.12	4.6.12
Detector-Dewar defects	3.6.1	4.6.1

4.5.2.2 Group B inspection. Group B inspection is not required.

4.5.2.3 Group C inspection. Group C inspection shall be conducted on Detector-Dewars selected from units which have passed the tests in 4.5.2.1. The tests of 4.6.9, 4.6.13, 4.6.15, and 4.7.5 shall be performed on a sample quantity.

4.5.2.3.1 Sampling for group C inspection. The tests of 4.6.9, 4.6.13, 4.6.15, and 4.7.5 shall be initially performed at a reduced sample rate of 3 percent of the specified lot (see 6.3.8) or one unit, whichever is the larger quantity.

a. When zero failure occurs, the 3 percent or one unit inspection sampling shall continue.

b. Failure of any unit to pass the inspection of 4.6.9, 4.6.13, 4.6.15, or 4.7.5 shall require 100 percent inspection of the entire lot for the failed attributes only with continued follow-on lots inspected until zero failure occurs in any lot. Upon achieving zero failure at 100 percent inspection, follow-on lot inspection shall be conducted at 10 percent sampling. Thereafter, upon achievement of zero failure during 10 percent sampling, the sample rate shall be lowered to 3 percent of the specified lot or one unit, whichever is larger. However, if one or more failures occur during the 10 percent inspection sampling, the 100 percent inspection testing shall be reinstituted.

4.5.2.3.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order (see 6.2).

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4.5.2.3.3 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to tests of table III.

4.5.2.4 Group D inspection. Group D inspection not required.

4.6 Test methods.

4.6.1 Detector-Dewar defects. The detector array shall be checked for defects. Failure to meet requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Time constant. Measurement shall be performed on 2 elements of the array each separated by at least 20 elements. The time constant is defined as the time required for the element response to reach 63 percent of its steady state value when irradiated by a 7.5 to 12 micrometer source which produces a signal-to-noise ratio not greater than 100:1. This time constant shall be considered equal to the time constant defined by the high frequency rolloff of the noise spectrum of the detector. Any measurement technique consistent with these definitions shall be considered acceptable. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Sensitive area. The physical area as defined by drawing 5004385 shall be determined by optical microscope examination. Measurement shall be made on 2 elements of the array each separated by at least 20 elements prior to the integration of the Dewar window with the Detector-Dewar. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Crosstalk. Measurements shall be made on 4 elements of the array. The crosstalk shall be measured under the following conditions:

a. An element shall be irradiated by a 7.5 to 12 micrometer source having a signal level 20 times greater than the noise level for the element. The image of the source shall fall entirely within the area of the detector element.

b. The signal level of the element adjacent to and on the same ground path of the irradiated element shall be measured. The element under test shall meet the requirements of 3.6.4.

Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Blackbody Detectivity. Noise measurements shall be made on each element of the array under the following conditions: detector cooled to 77 $\pm$ 3 K with LN_2, 295, $\pm$5 K background temperature and equivalent noise bandwidth of 9 kHz $\pm$ 200 Hz with low frequency 3dB point (high pass) no greater than 80 Hz. Using the results of these measurements, the results of the 500 K blackbody responsivity measurements, and the nominal detector size (see 6.3.7), the 500 K blackbody detectivity for each element of the array shall be calculated. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

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4.6.6 Detector resistance. The resistance of each element of the array shall be measured. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Responsivity. Blackbody responsivity of each element of the array shall be measured with the Dewar cooled with LN_2 to 77 $\pm$ 3 K. Nominal element size (see 6.3.7) shall be used in responsivity calculations and blackbody temperature shall be 500, $\pm$5 K. Failure to meet requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Temperature sensors. The 77 K calibration point shall be determined by cooling the inner Dewar with liquid nitrogen. 1.00 ma $\pm$0.01 ma bias shall be applied to the sensor diode and the voltage shall be monitored and recorded at the 77 K calibration point. Failure to meet requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Spectral response. The spectral response of the sixth element from each end of the array or adjacent element on the high end and the higher number of the two center elements shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. If these elements are defective an adjacent element may be substituted. The spectral sensitivity is defined as the 50 percent relative response points and shall be measured on the assembled Detector-Dewar. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Detector bias. The detector bias power shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. Failure to meet requirements of 3.6.10 shall constitute failure of this test.

4.6.11 Dewar warm and cold shields. During in-process inspection, the warm and cold shields shall be measured to ensure conformance with the requirement. Alignment of the cold shield relative to detector elements shall be checked 100 percent during in-process inspection. Failure to meet requirements of 3.6.11 shall constitute failure of this test.

4.6.12 Heat load. The heat load of the Detector-Dewar shall be determined with all detectors operating at specified bias. The heat load of the Detector-Dewar shall be determined by partially filling the Dewar's inner stem with liquid nitrogen and monitoring the rate of loss of the cryogen. The heat load shall be calculated from the average or integrated loss rate of the cryogen for the last 5 minutes of the Dewar's hold-time. For calculations, cooling produced by the nitrogen phase change only shall be considered as total useful energy. If this test is performed with the detector array unbiased, the results of this test shall be added to the results of the detector bias power test to determine the heat load. Failure to meet requirements of 3.6.12 shall constitute failure of this test.

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4.6.13 Vacuum integrity. Heat load shall be measured on the unbiased Detector-Dewar. The temperature of the Detector-Dewar shall then be elevated to +71°C for a period of 72 hours. At the completion of the temperature exposure and stabilization at ambient temperature, the heat load shall be measured again. Getters may not be fired during this test. Failure of the Detector-Dewar to meet the requirements of 3.6.12 (heat load) or a 15 percent increase in heat load between the measurement before and after temperature exposure shall constitute failure of this test.

4.6.14 Microphonics. Microphonics shall be measured while the Detector is vibrated in accordance with paragraph 4.6.14.1. The Detector-Dewar shall be cooled to 60-80K with full operating bias applied. Mounting to the cooler shall be made by tightening the mounting screws to a torque of 3.50 inch pounds minimum. Vibration excitation shall be measured by an accelerometer mounted on the mounting plate of the Detector-Dewar. The white noise reference level may be determined by averaging the noise in each channel from 12,000 to 20,000 Hz with vibration applied. Noise spectra shall be measured using a bandwidth between 50 Hz and 365 Hz. Failure of any channel to meet the requirements of 3.6.14 shall constitute a defect as defined in paragraph 3.6.1.

4.6.14.1 Microphonics vibration. The Detector-Dewar shall be coupled to a mechanical cryogenic refrigerator that produces random vibrational excitation of the Detector-Dewar in three mutually perpendicular axis. The frequency distribution of the vibrations shall be in a range from 0 to 20,000 Hertz. The RMS sum of the three axis vibration shall be at least 1.0 G RMS. Each axis shall produce a minimum vibrational excitation of 0.4 G RMS. In the frequency range of 2,000 to 20,000 Hz, the RMS sum shall be at least 0.8G RMS. The mechanical cryogenic refrigerator shall be mated to the Dewar with a metal bellows P/N SM-C-969732.

4.6.15 Apparent thermal mass (Cooldown). An HD-1132/UA cryogenic cooler shall be mated to the standard test mass (figure 3). The capacity of the HD-1132/UA shall be such that the 1.25 gram (250 joules) test mass is cooled from ambient (296K) temperature to 80K $\pm$ 1.5K in not less than 5 nor greater than 6 minutes. This measurement shall be made using a highly polished metal dewar without any radiation shielding. This test dewar and test mass shall consist of the following: a high vacuum of less than 10 microtorr, a 1.25 gram aluminum test mass, (see B2-2303020104 paragraph 4.2.1). The time to cool the thermal mass shall be recorded. The voltage output of the temperature sensor diodes in the Detector-Dewar under test shall be measured at 77 $\pm$ 1K and at ambient temperature. The voltage output at 80 K shall then be interpolated. The cooler shall then be coupled to the Detector using thermal grease and bellows. Operating bias power shall be applied and the time for the temperature of the diode to reach 80K shall be recorded. Failure of the diode to cool within the requirements of 3.6.15 shall constitute failure of this test.

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4.7 Environmental tests. The environmental tests shall be performed in accordance with MIL-STD-810 as modified by this specification. The performance tests required by MIL-STD-810 before and after each environmental test shall be those listed in table III. During the environmental test procedure, the units will not be energized or cooled down and will not be subjected to any performance monitoring.

4.7.1 Temperature shock. The Detector-Dewar shall be tested in accordance with MIL-STD-810, Method 503.1, Procedure I except that the temperature of step 2 shall be -54°C and the temperature of steps 1 and 4 shall be +71°C. Failure to meet performance parameters of Table III and damage criteria of 6.3.6 shall constitute failure of this test.

4.7.2 High temperature. The Detector-Dewar shall be tested in accordance with MIL-STD-810, Method 501.1, Procedure I. The temperature of steps 4 and 5 shall be +71°C. Failure to meet performance parameters of Table III and damage criteria of 6.3.6 shall constitute failure of this test.

4.7.3 Low temperature. The Detector-Dewar shall be tested in accordance with MIL-STD-810, Method 502.1, Procedure I except that the storage temperature of step 2 shall be -62°C and the low operating temperature of step 4 shall be -54°C. Failure to meet performance parameters of Table III and damage criteria of 6.3.6 shall constitute failure of this test.

4.7.4 Shock. The Detector-Dewar shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, figure 516.2-1 (100 g's peak value at 11 milliseconds time duration) and Procedure V. Failure to meet performance parameters of Table III and damage criteria of 6.3.6 shall constitute failure of this test.

4.7.5 Vibration. The Detector-Dewar shall be tested in accordance with MIL-STD-810, Method 514.2, Procedures I and VIII except that the curve shown in figure 1 of this specification shall be used in lieu of the MIL-STD-810 curves. The sinusoidal cycle time shall be 120 minutes/axis. The dwell time shall be 1/6 of the cycling time at each axis. Failure to meet performance parameters of Table III and damage criteria of 6.3.6 shall constitute failure of this test.

4.8 Reliability. A reliability prediction and math model shall be maintained by the contractor in accordance with MIL-STD-756 task sections 100 and 200 which reflect the current design and components of the Detector-Dewar. Failure of the prediction or model to explicitly delineate compliance with paragraph 3.8 shall constitute failure of the reliability requirement.

4.9 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

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4.9.1 First article inspection of packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.9.2 Quality conformance inspection of packaging.

4.9.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.9.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.9.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table IV, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE IV. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.10 Preconditioning. Each Detector-Dewar shall be subjected to preconditioning environments as shown in figure 2.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 2M-1-1-00-00-NU-X-XX-P-ED-A.

5.1.2 Level B. 2M-1-1-00-00-NU-X-XX-P-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The Detector-Dewar is intended for use in infrared systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required except for existing suppliers (see 6.3.9).

(1) Time frame for submission of first article test reports when first article tests are required.

(2) Time frame for approval of first article test reports when first article tests are required.

(3) First article reliability testing if required.

c. Production delivery schedule

(1) Defined in terms of monthly lots.

(2) Actions required relative to group C failures (see 4.5.2.3.2).

d. Level A or level B preservation and packaging (see section 5).

e. MIL-STD-810C shall be used for environmental tests of 4.7.

f. Detector-Dewar drawing numbers.

g. When First Article rough handling tests are not required.

h. Deliver all first article samples to - Commander, CECOM, Center for Night Vision and Electro-Optics, ATTN: AMSEL-RD-NV-TSD, Fort Belvoir, VA 22060-5677.

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i. Group C inspection monthly sample size to be established based on cumulative production totals of concurrent contracts.

j. MIL-STD-756B shall be used for reliability prediction of 4.8.

k. When First Article Packaging Inspection reports require acquisition activity approval prior to production unit packing.

6.3 Definitions.

6.3.1 Element. An element is defined as one channel in the 60-element Detector-Dewar.

6.3.2 Defective element. A defective element shall be defined as an element which fails to meet any one of the requirements of 3.6.5, 3.6.5.1, 3.6.6 and 3.6.7, 3.6.14.

6.3.3 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, shall be considered as a single defect. Elements whose D^* is less than 5.0×10^9 cm Hz $^{\frac{1}{2}}$ w^{-1} shall be opened prior to strapping.

6.3.4 Array axis. The array axis is defined as a line on the surface of the array passing through the center of the projected geometrical areas of each element.

6.3.5 Optical axis. The Detector-Dewar's optical axis is defined as a line perpendicular to the detector mounting plate that intersects the array axis at the midpoint of the array.

NOTE: One-half of the active detector elements are located on each side of the optical axis.

6.3.6 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of Detector-Dewar performance to values less than specified herein.

6.3.7 Nominal detector element size. Defined as 0.0016 by 0.0023 inch.

6.3.8 Specified lot quantity. The specified lot quantity shall be one-month's production unless the contract specifies a different lot basis such as one-week's production or those units processed as a single batch.

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6.3.9 Existing suppliers. Suppliers who are currently supplying equipment to this specification with a minimum average monthly rate of 10 units per month over the most current 12-month period. If a supplier cannot maintain the minimum monthly average, moves their production facilities, has significant management/personnel changes, and/or has significant changes in production methods-tooling-processes, a first article test (less reliability) performed annually shall sustain the existing supplier's status when performed on five (5) units manufactured by the supplier explicitly for this test.

6.4 Variation of responsivity with temperature. When coupled to a cryogenic refrigerator the temperature of the array may reach 55 K. Under those conditions the responsivity (paragraph 3.6.7) may decrease to as low as 5.6 X 10^3 volts/watt.

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Microphonic test caution. The microphonic requirements and tests specified herein are designed to be representative of an operational environment and combat scenerio for detectors mounted in a well designed system using current technology. If the system vibrational levels are increased above those specified, the microphonic noise may increase.

6.8 Subject term (keyword) Listing.

Dewar
Flir Systems
Sensors

6.9 Detector temperature. Detector temperature is defined as that indicated by the temperature sensor diode.

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR

Project 5855-A272

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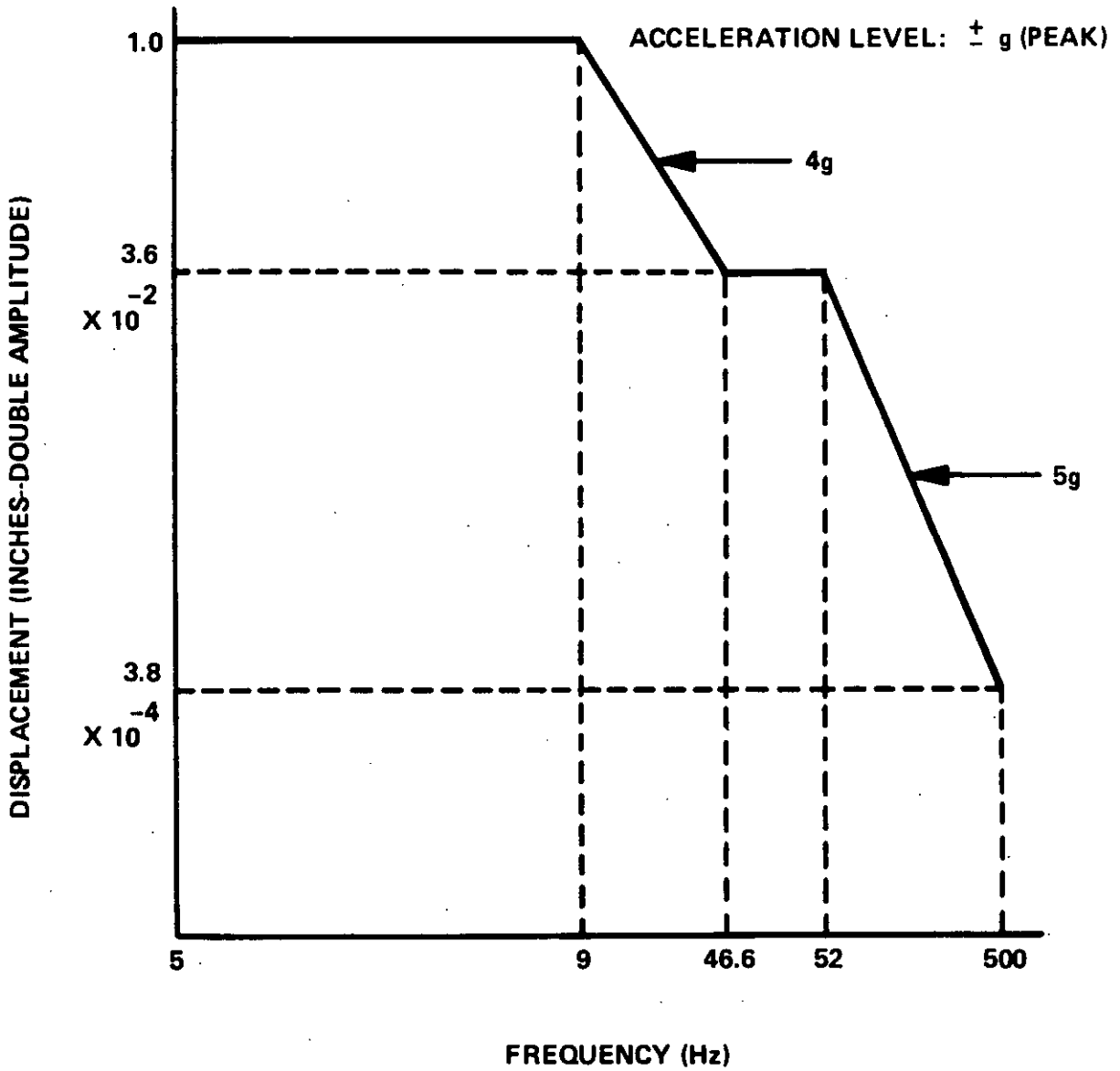
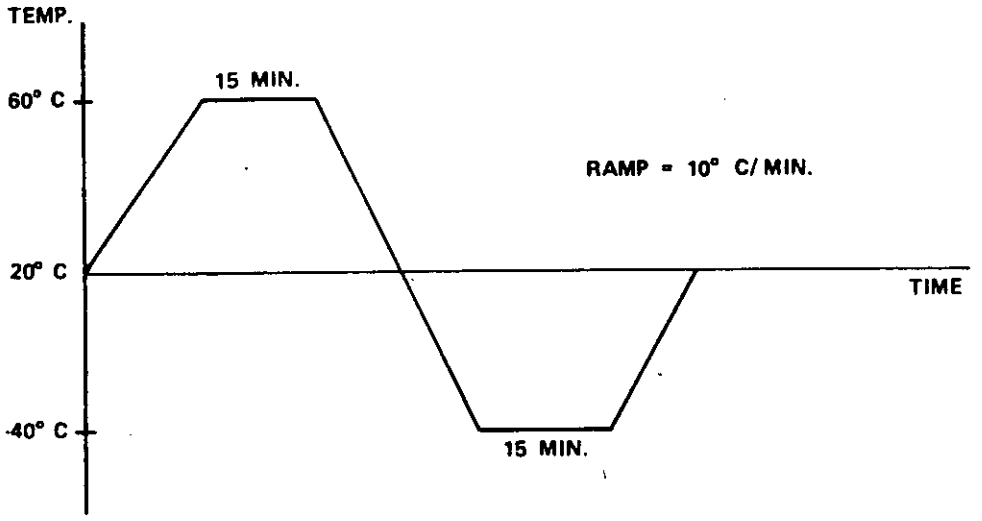


Figure 1. VIBRATION TEST PROFILE

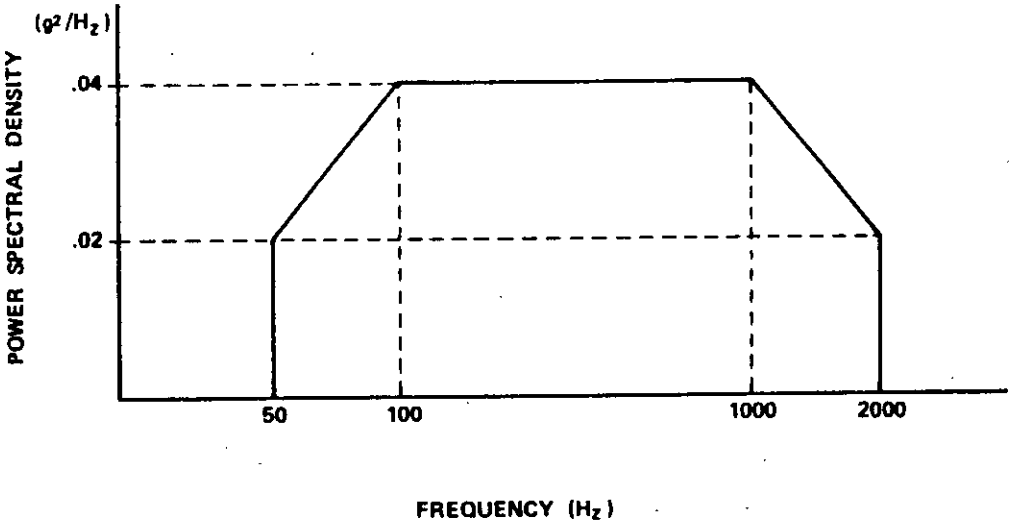
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TEMP CYCLE



VIBRATION CYCLE[1]

15 MIN



NOTES:
1. VIBRATION PERFORMED ALONG DETECTOR COLD WELL AXIS.
2. TEMP. CYCLE PRECEEDS VIBRATION CYCLE.

FIGURE 2. Preconditioning cycle

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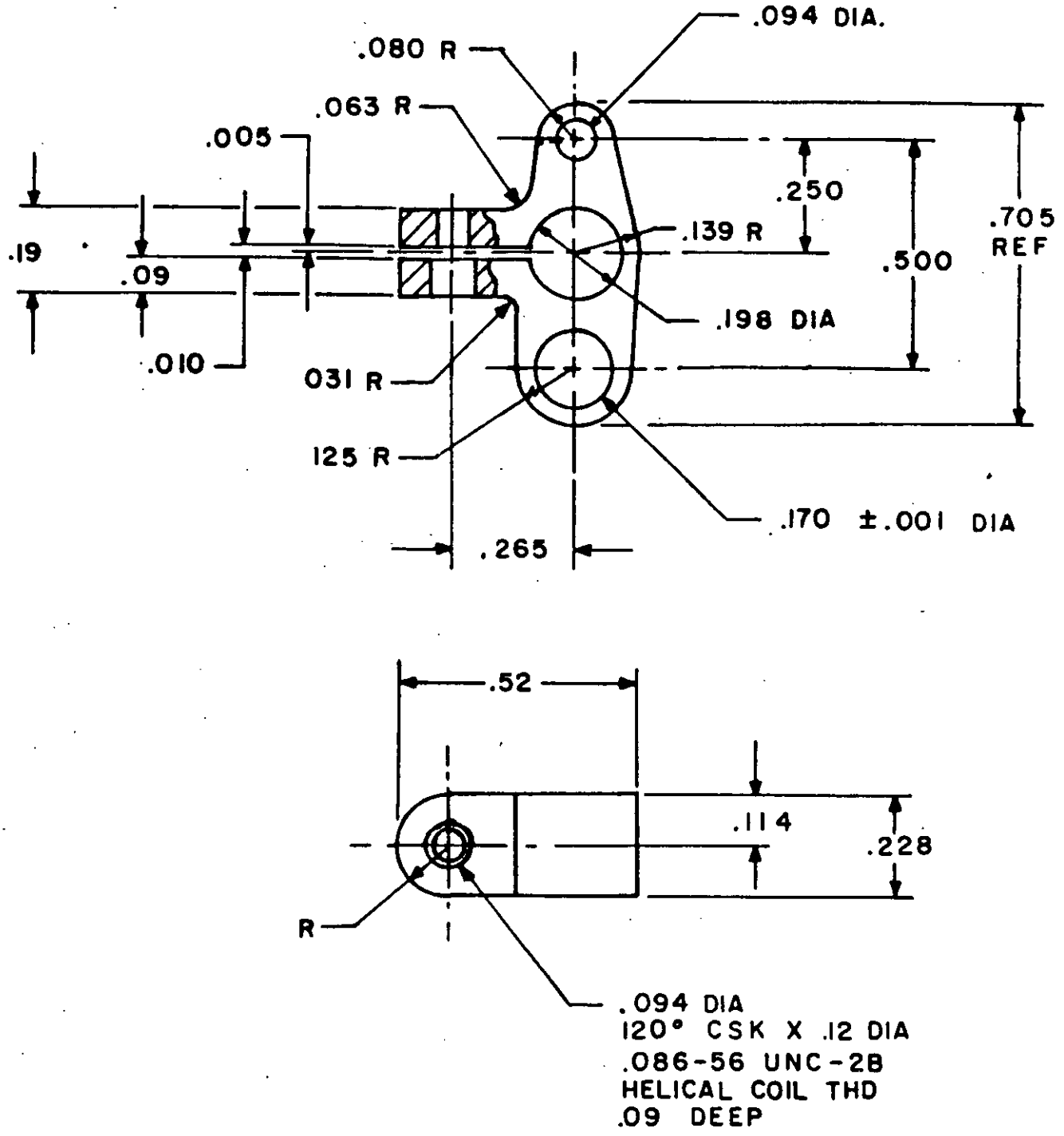


Figure 3. ALUMINUM BLOCK (250 JOULE MASS)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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